

Hanlin Zhu Paper Survey

Date: 2026-07-06. Scope: every current paper link crawled from Hanlin Zhu's homepage and publications page. Generated inside the Codex session using the local deepbrief-daily workflow as reference; the legacy DeepBrief engine was not invoked.

At a glance: 28 linked papers identified, 27 PDFs downloaded, 27 full-text extracts produced, 28 source-specific subagent read reports saved, 117 manifest records written, and 1 degraded source. Citation counts were looked up live through Semantic Scholar and OpenAlex; 18 papers have available counts and 10 are explicitly marked missing because the APIs blocked, throttled, or could not confidently match the paper.

Executive Synthesis

Hanlin Zhu's paper list has a clear center of gravity: reasoning is treated less as a prompt trick and more as a representational or environmental control problem. The most relevant cluster for an applied AI engineer is not the highest-cited older work; it is the agent/eval/runtime cluster around GSM-Agent, black-box API auditing, prompt caching, and latent/continuous reasoning methods 9 8 20 3.

The top DeepBrief pick is **GSM-Agent**. It wins on profile relevance because it turns agentic reasoning into a controllable missing-premise search environment, instruments traces with an agentic reasoning graph, and identifies "revisit after new information" as a concrete behavior to measure and encourage 9. Its citation count is missing, so this is a relevance-first choice, not a bibliometric one.

The citation leader is **Towards a Theoretical Understanding of the Reversal Curse via Training Dynamics**, with 36 Semantic Scholar citations in the live lookup 15. That paper is still useful, but its value for this reader is as a mechanism lens for directional sequence weights and relation inversion, not as the most directly actionable system/eval artifact.

Deep Dive Winner: GSM-Agent

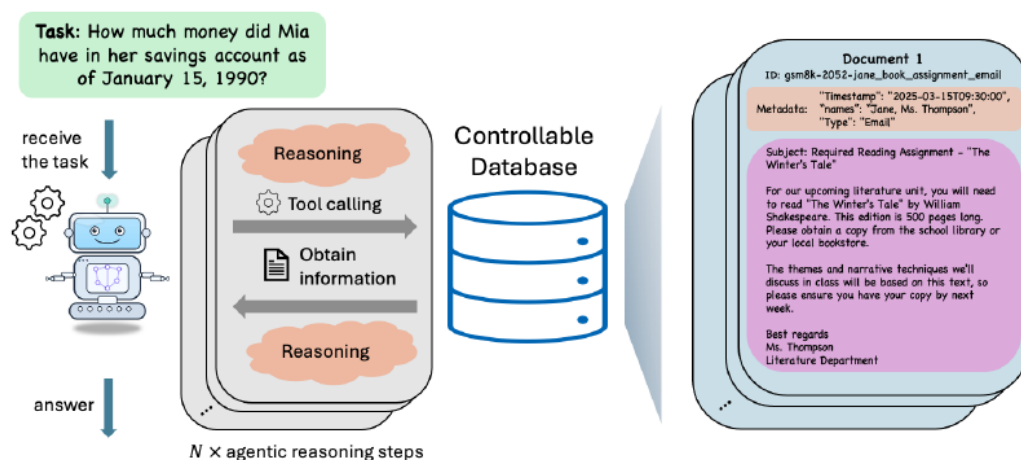


Figure 1. GSM-Agent turns a question-only task into an agent loop over a controllable document database; the useful audit signal is the path the agent takes while searching, revisiting, and deciding when to answer.

GSM-Agent converts GSM8K-style math problems into agentic search tasks by hiding the premises from the prompt, turning each premise into a searchable document, and giving the model tools such as Search and NextPage 9. The important move is not the math benchmark itself; it is the evaluation harness. The paper compares static reasoning against agentic reasoning in a controlled database, then maps tool traces onto a graph of exploration, exploitation, and revisit behavior 9.

The strongest operational lesson is that longer tool use is not enough. The subagent read report found the paper's most actionable behavioral motif in the "revisit" signal: strong agents return to earlier search regions after new information changes what they need 9. That is immediately useful for agent harness design because it can become a trace-level metric, a debugging lens, or a test-time tool affordance.

The caveat is equally important. The benchmark is synthetic and construction-heavy; documents, premise decomposition, filtering, and some verification steps depend on LLM-generated artifacts. Treat GSM-Agent as a clean harness for studying missing-premise search, not as a complete proxy for browser agents, mutable tools, permissions, adversarial sources, or long-running memory.

Relevance Ranking

Reader-profile relevance is scored against the local `profile.md` and `preferences.md`: reward agent mechanisms, evals, prompt/runtime mechanics, evidence trails, and implementation relevance; penalize thin announcements and low-applied theory. Citation rank is shown separately so older, well-cited theory does not crowd out currently useful engineering artifacts.

Priority	Citation rank	Paper	Relevance	Citation count	Why it ranks here
1	19	GSM-Agent	96	NA	Clean controllable agent benchmark, trace graph, and revisit behavior.
2	11	Auditing Black-Box LLM APIs with RUT	94	0	Directly relevant API integrity/eval mechanism.
3	5	Efficient Prompt Caching via Embedding Similarity	90	14	Production inference mechanism with concrete AUC/cache-efficiency evidence.
4	12	CopT	88	0	Continuous-prefix inference pattern for general and agentic reasoning.
5	8	Learning Personalized Alignment	85	4	Evaluator conditioning and preference-profile mechanics.
6	20	Token Assorted	82	NA	Latent-token reasoning compression with auditability tradeoff.
7	13	DiscoLoop	80	0	Looped latent/embedding handoff mechanism.
8	14	Reasoning by Superposition	78	0	Theoretical anchor for continuous reasoning as superposed search frontier.
9	4	Emergence of Superposition	77	17	Training-dynamics account for continuous thought.
10	3	Two-Hop Reasoning in Context	76	19	Mechanistic context-local binding under distractors.

Citation Ranking

Citation rank	Paper	Count	Source	Relevance note
1	Towards a Theoretical Understanding of the Reversal Curse	36	Semantic Scholar	Strong mechanism paper, less directly operational.
2	Guided Dialog Policy Learning	27	OpenAlex	Older reward-modeling/dialog-policy baseline.
3	How Do LLMs Perform Two-Hop Reasoning in Context?	19	Semantic Scholar	Useful mechanistic context-binding study.
4	Emergence of Superposition	17	Semantic Scholar	Important in the continuous-thought theory cluster.
5	Efficient Prompt Caching	14	Semantic Scholar	Highest-cited directly applied systems paper in the set.
6	Vector-Matrix-Vector Queries	8	OpenAlex	Theory-heavy query-complexity result.
7	Representation Complexity of RL	7	Semantic Scholar	Useful for RL representation tradeoffs.
8	Learning Personalized Alignment	4	OpenAlex	Relevant evaluator-pattern paper.

Citation rank	Paper	Count	Source	Relevance note
9	Safe Learning Under Irreversible Dynamics	3	Semantic Scholar	Safety theory with mentor/help assumptions.
10	Importance Weighted Actor-Critic	1	OpenAlex	Offline RL theory, narrow applied fit.

Paper Summaries

#	Paper	Relevance	Cites	Summary
1	DiscoLoop	80	0	Looped latent reasoning paper: discrete embeddings and continuous hidden states become useful only when hidden-state content is translated back into embedding geometry 1 .
2	Transformers Internalize Chain-of-Thought	74	0	Theory result showing how explicit parity-tree CoT can become layer-local hidden-state computation, best read as mechanism rather than deployable recipe 2 .
3	CopT	88	0	Answer-first, verify-with-continuous-prefixes, think-only-when-needed inference pattern with agentic benchmark evidence and logits/embedding-control caveats 3 .
4	Safe Learning Under Irreversible Dynamics	66	3	Reframes safe irreversible RL as active learning with mentor help and local generalization; strong formal result but depends on oracles/mentor access 4 .
5	Identity Bridge for Reversal Curse	72	0	Shows a format-sensitive OCR/identity bridge can partially repair reverse recall, but shortcut and tokenization limits remain 5 .
6	Multi-Objective Diffusion Learning	46	NA	Semi-supervised specialist-to-generalist pseudo-labeling theory for diffusion models; lower profile fit because it is not agent/eval/runtime focused 6 .
7	Emergence of Superposition	77	17	Training-dynamics account for how continuous CoT can represent multiple reasoning states, with synthetic evidence and theory constraints 7 .
8	RUT API Auditing	94	0	One-paid-call-per-prompt black-box model equality test using rank uniformity against a local reference model 8 .
9	GSM-Agent	96	NA	Controllable missing-premise agent benchmark; trace graph and revisit behavior are the main engineering takeaways 9 .
10	Two-Hop Reasoning in Context	76	19	Studies context-local binding and distractors; useful as a mechanistic lens for multi-hop failures rather than a broad LLM benchmark 10 .
11	Reasoning by Superposition	78	0	Theoretical case that continuous thoughts can carry a BFS-style frontier without discrete collapse 11 .

#	Paper	Relevance	Cites	Summary
12	Generalization or Hallucination?	75	NA	Uses out-of-context reasoning to separate generalization from hallucination, with OCR and factorized attention as the core mechanism 12 .
13	Token Assorted	82	NA	Mixes latent and text tokens for shorter reasoning traces, improving efficiency while creating an auditability tradeoff 13 .
14	Avoiding Catastrophe by Asking for Help	68	NA	Formal online-learning safety result: occasional mentor help is insufficient without structure; local generalization is the positive path 14 .
15	Reversal Curse Training Dynamics	70	36	Citation leader; the crisp message is that CE-trained autoregressive models learn directional sequence weights, not relation algebra 15 .
16	Personalized Alignment Evaluator	85	4	Conditions an evaluator on a small reviewer profile; high relevance for preference-aware evals and personalized LLM-as-judge workflows 16 .
17	Starling-7B RLAIIF	30	NA	Degraded: site listing and OpenReview URL verified, but OpenReview PDF/API access was blocked; no mechanism summary included 17 .
18	Statistical Watermarking	64	NA	Strong theoretical watermarking anchor, unifying statistical watermarks as hypothesis tests; practical robust model-agnostic deployment remains open 18 .
19	Representation Complexity in RL	56	7	Separates model-based and model-free representation complexity; useful for RL theory, less direct for applied agent systems 19 .
20	Efficient Prompt Caching	90	14	Learns response-equivalence embeddings for prompt caching; reports AUC 0.51 to 0.81 and best streaming efficiency 46.0% to 54.0% 20 .
21	End-to-End Story Plot Generator	54	NA	Distills prompt-engineered hierarchical story planning into a one-call Llama2 plot generator; evidence is GPT-4 judged and plot-only 21 .
22	Offline Goal-Conditioned RL	52	NA	Theory anchor for offline GCRL with regularized occupancy and dual-V learning; strong assumptions and slow rate limit applied fit 22 .
23	Importance Weighted Actor-Critic	49	1	MIS-weighted average-Bellman-error actor-critic for conservative offline RL; clean theory with narrow empirical validation 23 .
24	Optimal Conservative Offline RL	48	0	Shows conservatism comes from bounded occupancy ratios, with augmented Lagrangian supplying validity rather than pessimism 24 .

#	Paper	Relevance	Cites	Summary
25	Surprise-Bound RL	50	0	Distribution-dependent surprise bound and optimistic LSVI with doubling epochs; theoretical, no experiments, weak citation metadata 25 .
26	Average-Case Communication Complexity	40	NA	Distribution-preserving average-case communication-complexity toolkit for planted-structure lower bounds 26 .
27	Vector-Matrix-Vector Queries	42	8	Unifying bilinear-query framework across linear algebra, statistics, and graph problems; theory-only for this reader 27 .
28	Guided Dialog Policy Learning	58	27	Early reward-modeling bridge from terminal success to per-turn learned evaluators for multi-domain dialog policy 28 .

Figures and Mechanisms

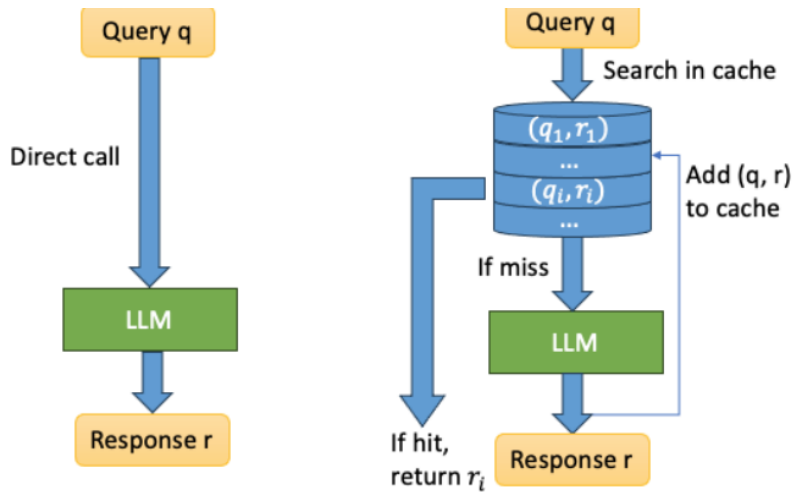


Figure 2. Prompt caching compares a direct LLM call with a cache path that reuses a prior prompt-response pair when the new prompt can share the same response.

Prompt caching is the most immediately product-adjacent systems paper in the corpus. The mechanism is not generic semantic similarity; it is response equivalence: can two prompts be answered by the same cached response? The paper constructs a hard Natural Questions pair dataset, labels pairs with GPT-4, and fine-tunes embeddings to distinguish cache hits from misses [20](#).

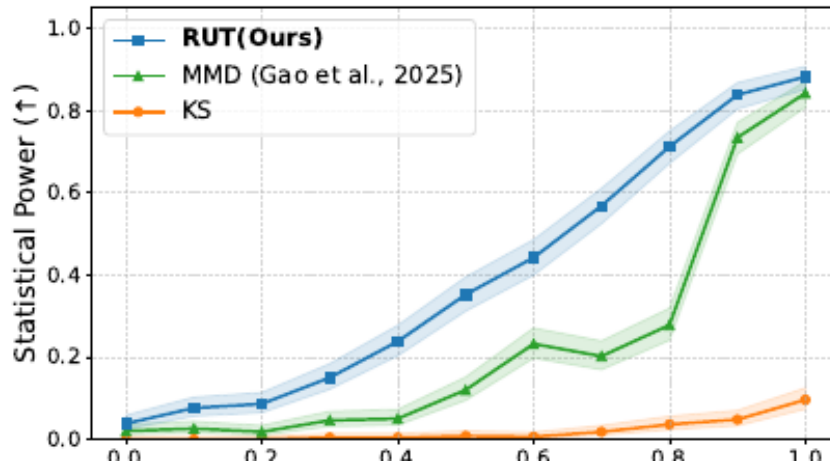


Figure 3. RUT detects substitutions by testing whether target API outputs have uniform rank statistics under the claimed local reference model.

RUT is a compact example of the profile's preferred evidence style: a clear threat model, a test statistic, and a constrained operational budget. It assumes a faithful local reference model, asks the remote API for one completion per prompt, samples local reference completions, ranks the target score under the reference distribution, and applies a uniformity test 8.

Condensed Evidence Matrix

Full local matrix: [artifacts/hanlin-zhu-paper-survey-2026-07-06/verification/evidence-matrix.md](#).

Source	Evidence-backed claim	Local audit anchor	Status
9	GSM-Agent is a controllable missing-premise search benchmark whose trace graph makes revisit behavior visible.	reviews/subagents/read-hz09-...md	Verified
8	RUT audits open-weight LLM APIs through randomized rank uniformity under a local reference distribution.	reviews/subagents/read-hz08-...md	Verified
20	Prompt-cache embeddings should model response equivalence, not only semantic similarity.	reviews/subagents/read-hz20-...md	Verified
3	CopT uses continuous prefixes to verify draft answers and allocate extra thinking where needed.	reviews/subagents/read-hz03-...md	Verified
16	PerSE conditions an evaluator on reviewer profiles and outputs personalized scores/explanations.	reviews/subagents/read-hz16-...md	Verified
13	Token Assorted compresses reasoning with latent/text-token mixtures while reducing trace transparency.	reviews/subagents/read-hz13-...md	Verified
15	Reversal-curse theory says directional autoregressive training does not automatically learn inverse relation algebra.	reviews/subagents/read-hz15-...md	Verified
17	Starling-7B is listed on the site, but full-text OpenReview access was blocked in this run.	reviews/subagents/read-hz17-...md	Degraded

Manifest and Verification Artifacts

Artifact	Path	Count / status
Candidate log	artifacts/hanlin-zhu-paper-survey-2026-07-06/sources/candidates.jsonl	28 paper records
Ranking log	artifacts/hanlin-zhu-paper-survey-2026-07-06/sources/ranking.jsonl	28 ranked records

Artifact	Path	Count / status
Raw artifact manifest	artifacts/hanlin-zhu-paper-survey-2026-07-06/sources/manifest.jsonl	117 records; 112 downloaded, 1 degraded, 4 blocked
Paper PDFs	artifacts/hanlin-zhu-paper-survey-2026-07-06/sources/papers/	27 PDFs, 27 text extracts
Figure crops	artifacts/hanlin-zhu-paper-survey-2026-07-06/images/figures/	3 cropped source figures
Subagent reports	artifacts/hanlin-zhu-paper-survey-2026-07-06/reviews/subagents/	28 reports, one per linked paper
Fanout report	artifacts/hanlin-zhu-paper-survey-2026-07-06/reviews/fanout-report.md	5 completed waves
Evidence matrix	artifacts/hanlin-zhu-paper-survey-2026-07-06/verification/evidence-matrix.md	28 paper rows

Verification summary: the site crawl identified 28 current paper links on the publications page; the home page did not add a distinct current paper beyond that set. ArXiv PDFs were downloaded and text-extracted for 27 papers. OpenReview blocked Starling-7B PDF/attachment/API access with 403 or challenge responses, so that source is degraded and not used for mechanism claims. Citation counts are live API values as of 2026-07-06 and should not be compared to Google Scholar counts.

Residual Risks

- Citation counts are incomplete for 10 papers because Semantic Scholar/OpenAlex throttled, blocked, or failed to confidently match them. Missing counts are not treated as zero.
- Several papers are future-dated or 2026 preprints, so current citation counts are unstable and likely understate eventual influence.
- Source-specific reads used local PDF text extraction and subagent reports; no downloaded code repositories were executed.
- Starling-7B remains degraded until OpenReview full text can be fetched through a browser-authenticated or otherwise permitted path.

Citation Appendix

Source 1: DiscoLoop: Looping Discrete Embeddings and Continuous Hidden States for Multi-hop Reasoning

- URL: <https://arxiv.org/abs/2607.00341>
- Type: paper

Source 2: Transformers Provably Learn to Internalize Chain-of-Thought

- URL: <https://arxiv.org/abs/2605.28600>
- Type: paper

Source 3: CopT: Contrastive On-Policy Thinking with Continuous Spaces for General and Agentic Reasoning

- URL: <https://arxiv.org/abs/2605.20075>
- Type: paper

Source 4: Safe Learning Under Irreversible Dynamics via Asking for Help

- URL: <https://arxiv.org/abs/2502.14043>
- Type: paper

Source 5: Breaking the Reversal Curse in Autoregressive Language Models via Identity Bridge

- URL: <https://arxiv.org/abs/2602.02470>
- Type: paper

Source 6: Multi-Objective Learning for Diffusion Models: A Statistical Theory under Semi-Supervised Learning

- URL: <https://arxiv.org/abs/2605.25210>
- Type: paper

Source 7: Emergence of Superposition: Unveiling the Training Dynamics of Chain of Continuous Thought

- URL: <https://arxiv.org/abs/2509.23365>
- Type: paper

Source 8: Auditing Black-Box LLM APIs with a Rank-Based Uniformity Test

- URL: <https://arxiv.org/abs/2506.06975>
- Type: paper

Source 9: GSM-Agent: Understanding Agentic Reasoning Using Controllable Environments

- URL: <https://arxiv.org/abs/2509.21998>

- Type: paper

Source 10: How Do LLMs Perform Two-Hop Reasoning in Context?

- URL: <https://arxiv.org/abs/2502.13913>
- Type: paper

Source 11: Reasoning by Superposition: A Theoretical Perspective on Chain of Continuous Thought

- URL: <https://arxiv.org/abs/2505.12514>
- Type: paper

Source 12: Generalization or Hallucination? Understanding Out-of-Context Reasoning in Transformers

- URL: <https://arxiv.org/abs/2506.10887>
- Type: paper

Source 13: Token Assorted: Mixing Latent and Text Tokens for Improved Language Model Reasoning

- URL: <https://arxiv.org/abs/2502.03275>
- Type: paper

Source 14: Avoiding Catastrophe in Online Learning by Asking for Help

- URL: <https://arxiv.org/abs/2402.08062>
- Type: paper

Source 15: Towards a Theoretical Understanding of the Reversal Curse via Training Dynamics

- URL: <https://arxiv.org/abs/2405.04669>
- Type: paper

Source 16: Learning Personalized Alignment for Evaluating Open-ended Text Generation

- URL: <https://arxiv.org/abs/2310.03304>
- Type: paper

Source 17: Starling-7B: Improving Helpfulness and Harmlessness with RLAI

- URL: <https://openreview.net/forum?id=GqDntYTTbk#discussion>
- Type: paper listing; degraded full-text access

Source 18: Towards Optimal Statistical Watermarking

- URL: <https://arxiv.org/abs/2312.07930>
- Type: paper

Source 19: On Representation Complexity of Model-based and Model-free Reinforcement Learning

- URL: <https://arxiv.org/abs/2310.01706>
- Type: paper

Source 20: Efficient Prompt Caching via Embedding Similarity

- URL: <https://arxiv.org/abs/2402.01173>
- Type: paper

Source 21: End-to-end Story Plot Generator

- URL: <https://arxiv.org/abs/2310.08796>
- Type: paper

Source 22: Provably Efficient Offline Goal-Conditioned Reinforcement Learning with General Function Approximation and Single-Policy Concentrability

- URL: <https://arxiv.org/abs/2302.03770>
- Type: paper

Source 23: Importance Weighted Actor-Critic for Optimal Conservative Offline Reinforcement Learning

- URL: <https://arxiv.org/abs/2301.12714>
- Type: paper

Source 24: Optimal Conservative Offline RL with General Function Approximation via Augmented Lagrangian

- URL: <https://arxiv.org/abs/2211.00716>
- Type: paper

Source 25: Provably Efficient Reinforcement Learning via Surprise Bound

- URL: <https://arxiv.org/abs/2302.11634>
- Type: paper

Source 26: Average-Case Communication Complexity of Statistical Problems

- URL: <https://arxiv.org/abs/2107.01335>
- Type: paper

Source 27: Vector-Matrix-Vector Queries for Solving Linear Algebra, Statistics, and Graph Problems

- URL: <https://arxiv.org/abs/2006.14015>
- Type: paper

Source 28: Guided Dialog Policy Learning: Reward Estimation for Multi-Domain Task-Oriented Dialog

- URL: <https://arxiv.org/abs/1908.10719>
- Type: paper